

Package ‘LRg’

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Title Logistic Regression

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Description Functions for fitting different logistic regression models. Models included are the multi-nominal logistic regression model, the parallel adjacent-categories logit model, and multi-dimensional stereotype ordered and unordered regression models.

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auto_factor	<i>Automatic recoding into factors</i>
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Description

Automatic recoding into factors

Usage

```
auto_factor(formula, data, max_levels = 10)
```

Arguments

formula	model expression
data	data frame
max_levels	maximum number of factor levels

Value

recoded data frame

btls	<i>Backtracking line search</i>
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Description

Backtracking line search

Usage

```
btls(x, X, Z, dx, u, alpha = 1, rho = 0.5, c = 1e-04, maxit = 20)
```

Arguments

x	parameter vector
X	design matrix
Z	response dummies
dx	search direction
u	score vector
alpha	step size
rho	contraction factor
c	Armijo parameter
maxit	maximum number of iterations

Value

new parameter vector

btlS_stn	<i>Backtracking line search</i>
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Description

Backtracking line search

Usage

```
btlS_stn(x, X, Z, q, dx, u, alpha = 1, rho = 0.5, c = 1e-04, maxit = 20)
```

Arguments

x	parameter vector
X	design matrix
Z	response dummies
q	rank
dx	search direction
u	score vector
alpha	step size
rho	contraction factor
c	Armijo parameter
maxit	maximum number of iterations

Value

new parameter vector

cum_contr	<i>Cumulative binary recoding</i>
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Description

Cumulative binary recoding

Usage

```
cum_contr(x)
```

Arguments

x	ordered factor
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Value

cumulative contrast

eyedisease

Empirical dataset

Description

The data are 720 observations on 6 variables.

Usage

eyedisease

Format

A data frame with 720 rows and 6 columns

diab diabetes

gh glycosylated haemoglobin level

dbp diastolic blood pressure

prot proteinuria

merl maximum severity of retinopathy in the left and right eye

Source

From the Wisconsin Epidemiological Study of Diabetic Retinopathy (Klein et al., 1984). From R package ordinalgmifs (Archer et al., 2014).

jacob

Jacobian

Description

Jacobian

Usage

jacob(alpha, gamma, Lambda, Phi)

Arguments

alpha vector of intercepts

gamma vector of regressor effects

Lambda matrix of regressor effects

Phi matrix of weights

Value

Jacobian matrix

jacob_stn	<i>Jacobian</i>
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Description

Jacobian

Usage

jacob_stn(Phi, Gamma)

Arguments

Phi	matrix
Gamma	matrix

Value

Jacobian

logLik	<i>The log-likelihood function</i>
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Description

The log-likelihood function

Usage

logLik(x, X, Z)

Arguments

x	parameter vector
X	desgin matrix
Z	response dummies

Value

log-likelihood value

logLik_stn	<i>The log-likelihood function</i>
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Description

The log-likelihood function

Usage

```
logLik_stn(x, X, Z, q)
```

Arguments

x	parameter vector
X	design matrix
Z	response dummies
q	rank

Value

log-likelihood value

lrt	<i>Likelihood ratio test</i>
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Description

Likelihood ratio test

Usage

```
lrt(obj1, obj2)
```

Arguments

obj1	fit results model 1
obj2	fit results model 2

Value

test statistic, df, and p-value

mlr	<i>Multinomial logistic regression</i>
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Description

This function can be used to fit the multinomial logistic regression model.

Usage

```
mlr(formula, data, maxit = 100)
```

Arguments

formula	model expression
data	data frame
maxit	maximum number of iterations

Value

A list of

coefficients	Parameter estimates
fitted.values	The model implied probabilities
residuals	Dummy responses minus fitted values
pred.cat	The model implied categories

Author(s)

David J. Hessen

olr	<i>Ordinal logistic regression</i>
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Description

This function can be used to fit the full rank ordinal logistic regression model using projected iterative weighted least squares.

Usage

```
olr(formula, data, maxit = 100)
```

Arguments

formula	model expression
data	data frame
maxit	maximum number of iterations

Value

A list of

<code>coefficients</code>	Parameter estimates
<code>fitted.values</code>	The model implied probabilities
<code>residuals</code>	Dummy responses minus fitted values
<code>pred.cat</code>	The model implied categories

Author(s)

David J. Hessen

<code>pac</code>	<i>Parallel adjacent-categories logit model</i>
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Description

This function can be used to fit the parallel adjacent-categories logit model.

Usage

```
pac(formula, data, maxit = 100)
```

Arguments

<code>formula</code>	model expression
<code>data</code>	data frame
<code>maxit</code>	maximum number of iterations

Value

A list of

<code>coefficients</code>	Parameter estimates
<code>fitted.values</code>	The model implied probabilities
<code>residuals</code>	Dummy responses minus fitted values
<code>pred.cat</code>	The model implied categories

Author(s)

David J. Hessen

`sim_data`*Example dataset*

Description

A data set containing a single ordinal response variable and four continuous regressor variables. The response variable has four levels.

Usage`sim_data`**Format**

A data frame with 300 rows and 5 columns

y Response

X1 Regressor variable 1

X2 Regressor variable 2

X3 Regressor variable 3

X4 Regressor variable 4

Source

Simulated data

`solve_safe`*Solving Idx=u*

Description

Solving Idx=u

Usage`solve_safe(I, u, lambda = 1e-08)`**Arguments**

`I` information matrix or Hessian

`u` score or gradient

`lambda` ridge penalty parameter

Value

ridge information matrix or Hessian

stn	<i>Stereotype nominal regression</i>
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Description

This function can be used to fit a q-dimensional stereotype nominal regression model.

Usage

```
stn(formula, q = NULL, data, stv = NULL, csv = 0.001, tol = 1e-08, maxit = 100)
```

Arguments

formula	model expression
q	rank of the matrix of regressor effects. If not supplied, then it is set to its maximum.
data	data frame
stv	vector of starting values
csv	common starting value
tol	convergence tolerance
maxit	maximum number of iterations

Value

A list of the following components

alpha	vector of estimated intercepts
Beta	matrix of estimated regressor effects
Gamma, Phi	matrices of parameter estimates
loglik	value of the log-likelihood function
npars	number of estimated parameters
AIC	Akaike information criterion
BIC	Bayesian information criterion
fitted.values	model implied probabilities
residuals	dummy responses minus fitted values
pred.cat	model implied categories
obs.cat	observed categories
CS, NK, MF	Cox and Snell, Nagelkerke, and McFadden pseudo-R-squared values

Author(s)

David J. Hessen

sto	<i>Stereotype ordered regression</i>
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Description

This function can be used to fit a q-dimensional stereotype ordered regression model.

Usage

```
sto(formula, q = NULL, data, tol = 1e-08, maxit = 100)
```

Arguments

formula	model expression
q	rank of the matrix of regressor effects. If not supplied, then it is set to its maximum.
data	data frame
tol	convergence tolerance
maxit	maximum number of iterations

Value

A list of the following components

alpha	vector of estimated intercepts
Beta	matrix of estimated regressor effects
Gamma, Phi	matrices of parameter estimates
loglik	Value of the log-likelihood function
npars	Number of estimated parameters
AIC	Akaike information criterion
BIC	Bayesian information criterion
CS, NK, MF	Cox and Snell, Nagelkerke, and McFadden pseudo-R-squared values

Author(s)

David J. Hessen

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